NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES
(Peshawar CAMPUS)
FAST School of Computing
Spring 2025

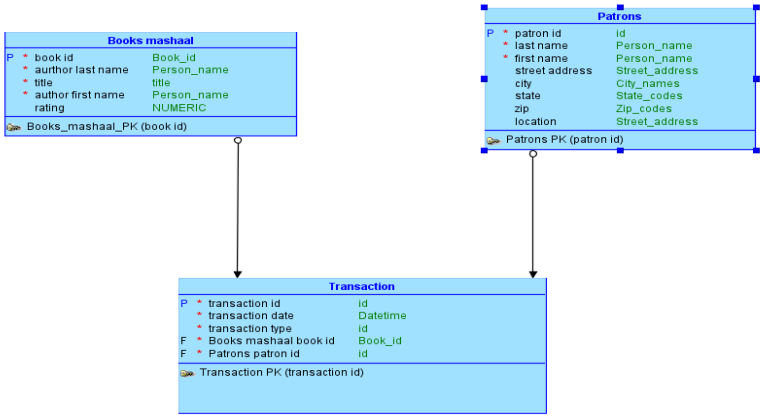
Name : Mashaal Ishtiaq

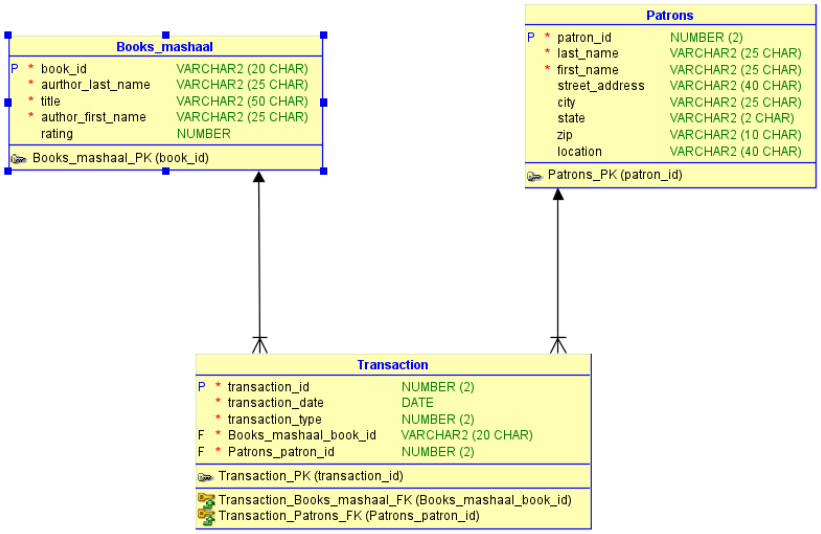
Roll# : 22p-9326

Section : BCS-6B

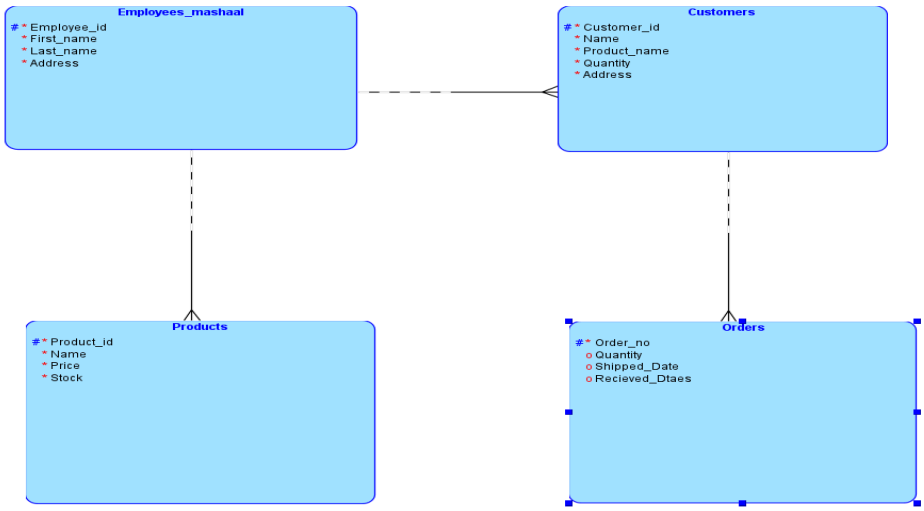
LAB TASK # 8

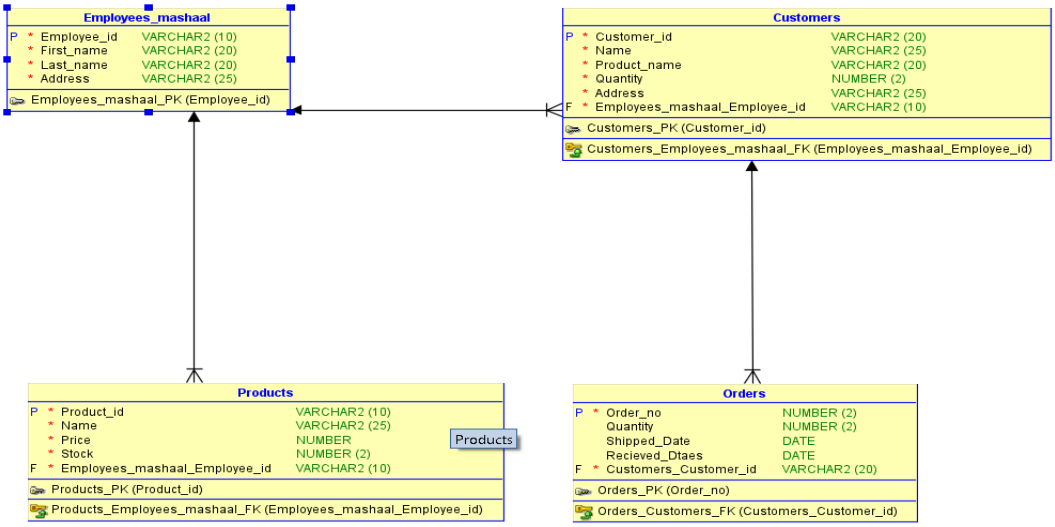
TASK#1



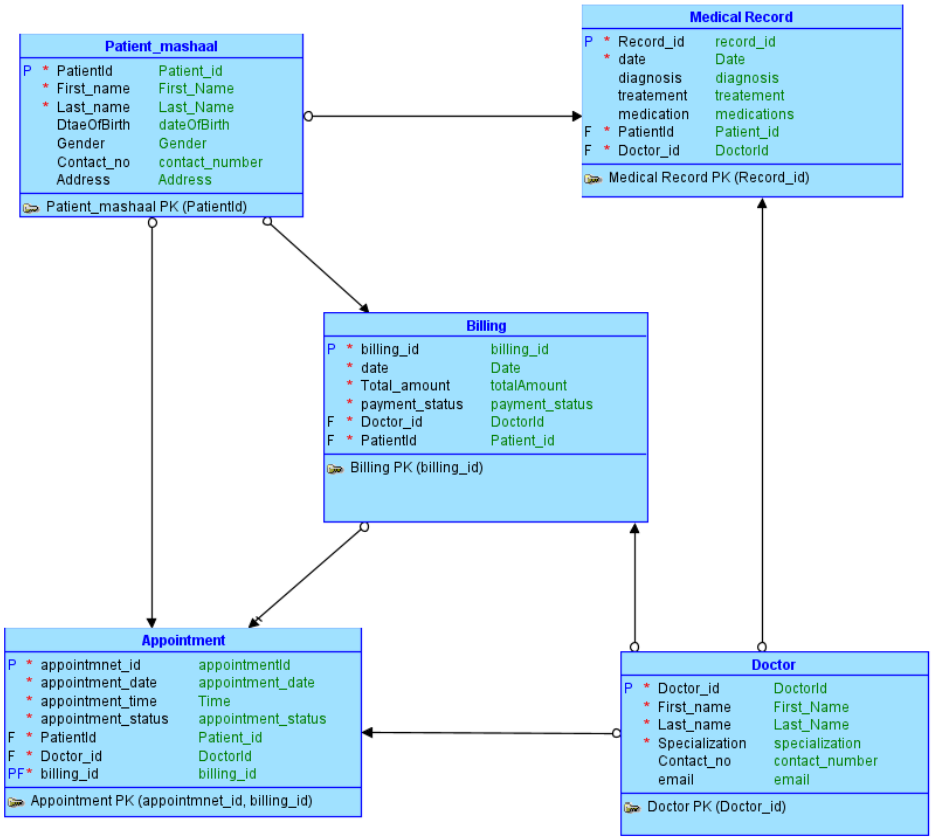

Books_mashaal
P * book_id VARCHAR2 (20 CHAR)
* aurthor_last_name VARCHAR2 (25 CHAR)
* title VARCHAR2 (50 CHAR)
* author_first_name VARCHAR2 (25 CHAR)
rating NUMBER
Books_mashaal_PK (book_id)
Patrons
P * patron_id NUMBER (2)
* last_name VARCHAR2 (25 CHAR)
* first_name VARCHAR2 (25 CHAR)
street_address VARCHAR2 (40 CHAR)
city VARCHAR2 (25 CHAR)
state VARCHAR2 (2 CHAR)
zip VARCHAR2 (10 CHAR)
location VARCHAR2 (40 CHAR)
Patrons_PK (patron_id)
Transaction
P * transaction_id NUMBER (2)
* transaction_date DATE
* transaction_type NUMBER (2)
F * Books_mashaal_book_id VARCHAR2 (20 CHAR)
F * Patrons_patron_id NUMBER (2)
Transaction_PK (transaction_id)
Transaction_Books_mashaal_FK (Books_mashaal_book_id)
Transaction_Patrons_FK (Patrons_patron_id)

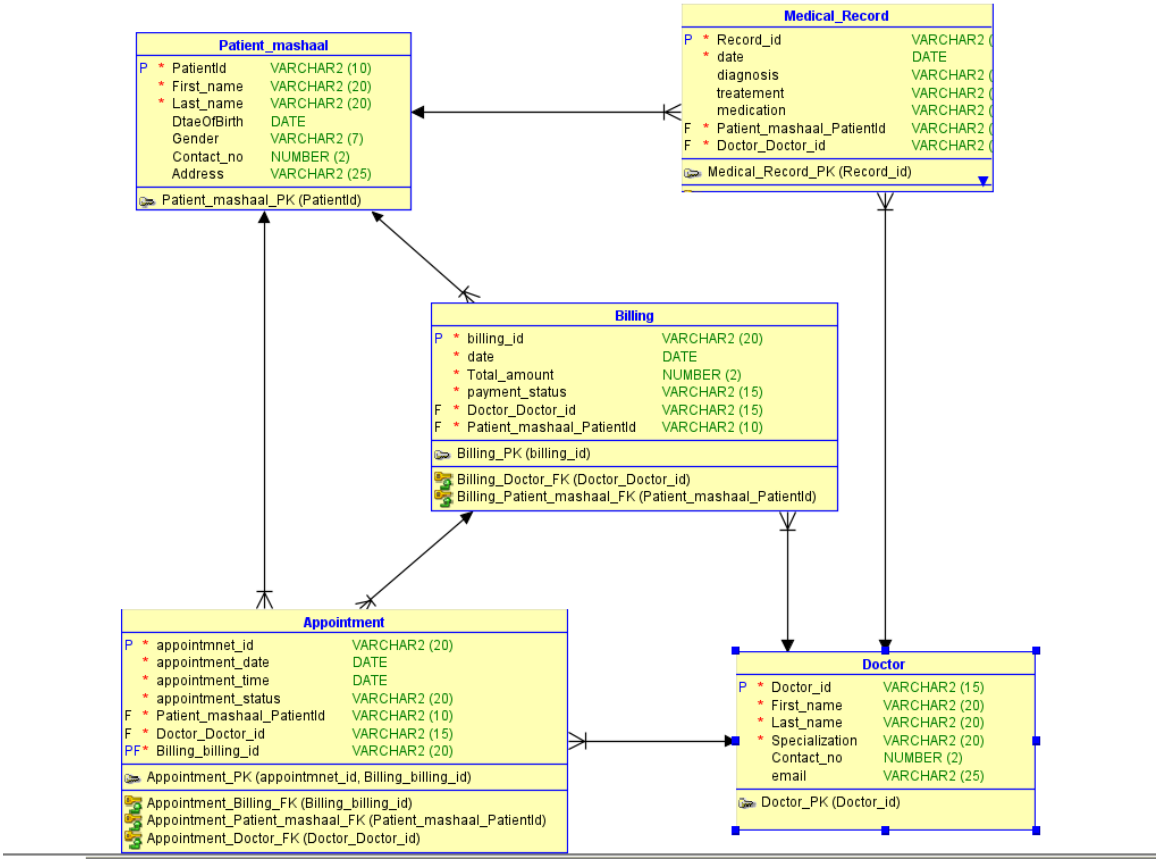
TASK#2


Employees_mashaal
* Employee_id
* First_name
* Last_name
* Address
Customers
* Customer_id
* Name
* Product_name
* Quantity
* Address
Products
* Product_id
* Name
* Price
* Stock
Orders
* Order_no
o Quantity
o Shipped_Date
o Recieved_Dtaes

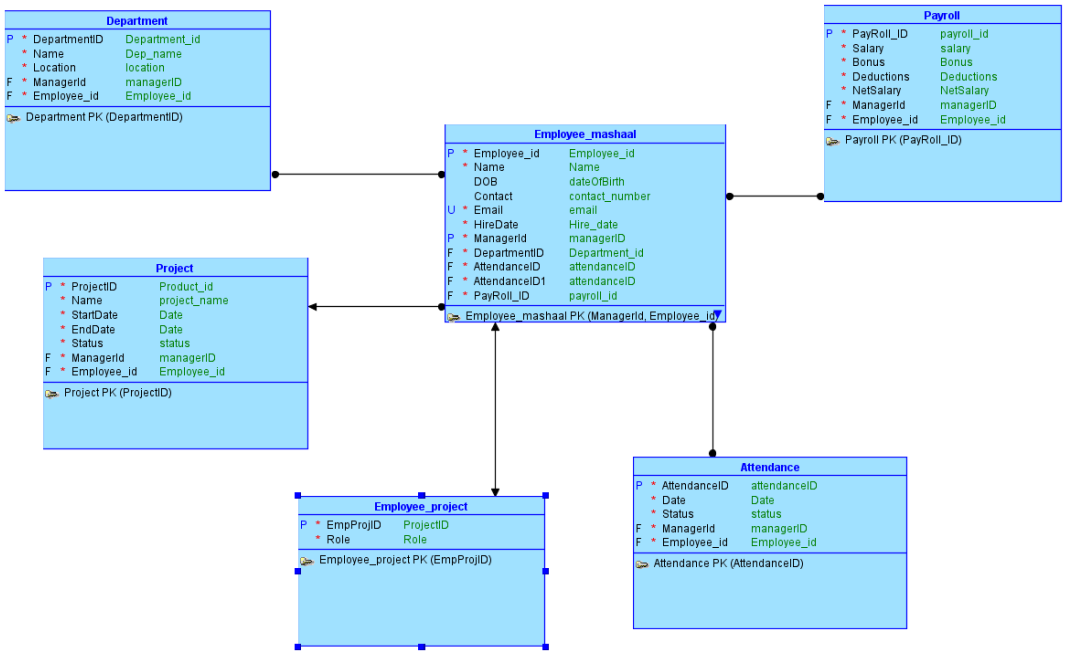

Employees_mashaal
P * Employee_id VARCHAR2 (10)
* First_name VARCHAR2 (20)
* Last_name VARCHAR2 (20)
* Address VARCHAR2 (25)
Employees_mashaal_PK (Employee_id)
Customers
P * Customer_id VARCHAR2 (20)
* Name VARCHAR2 (25)
* Product_name VARCHAR2 (20)
* Quantity NUMBER (2)
* Address VARCHAR2 (25)
F * Employees_mashaal_Employee_id VARCHAR2 (10)
Customers_PK (Customer_id)
Customers_Employees_mashaal_FK (Employees_mashaal_Employee_id)
Products
P * Product_id VARCHAR2 (10)
* Name VARCHAR2 (25)
* Price NUMBER
* Stock NUMBER (2)
F * Employees_mashaal_Employee_id VARCHAR2 (10)
Products_PK (Product_id)
Products_Employees_mashaal_FK (Employees_mashaal_Employee_id)
Products
Orders
P * Order_no NUMBER (2)
Quantity NUMBER (2)
Shipped_Date DATE
Recieved_Dtaes DATE
F * Customers_Customer_id VARCHAR2 (20)
Orders_PK (Order_no)
Orders_Customers_FK (Customers_Customer_id)

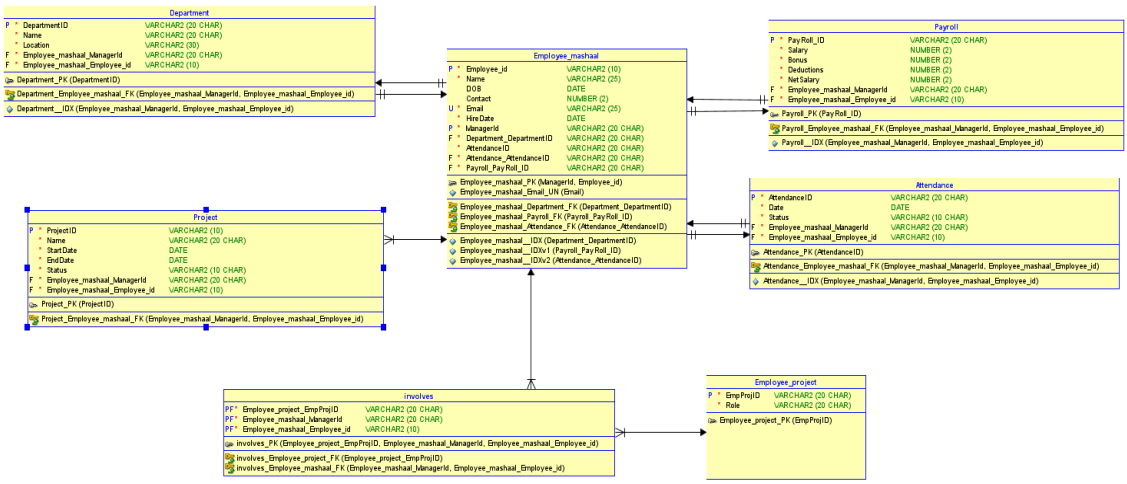
TASK#3


Patient_mashaal
P * PatientId Patient_id
* First_name First_Name
* Last_name Last_Name
DtaeOfBirth dateOfBirth
Gender Gender
Contact_no contact_number
Address Address
Patient_mashaal PK (PatientId)
Medical Record
P * Record_id record_id
* date Date
diagnosis diagnosis
treatement treatement
medication medications
F * PatientId Patient_id
F * Doctor_id DoctorId
Medical Record PK (Record_id)
Billing
P * billing_id billing_id
* date Date
* Total_amount totalAmount
* payment_status payment_status
F * Doctor_id DoctorId
F * PatientId Patient_id
Billing PK (billing_id)
Appointment
P * appointmnet_id appointmentId
* appointment_date appointment_date
* appointment_time Time
* appointment_status appointment_status
F * PatientId Patient_id
F * Doctor_id DoctorId
PF * billing_id billing_id
Appointment PK (appointmnet_id, billing_id)
Doctor
P * Doctor_id DoctorId
* First_name First_Name
* Last_name Last_Name
* Specialization specialization
Contact_no contact_number
email email
Doctor PK (Doctor_id)

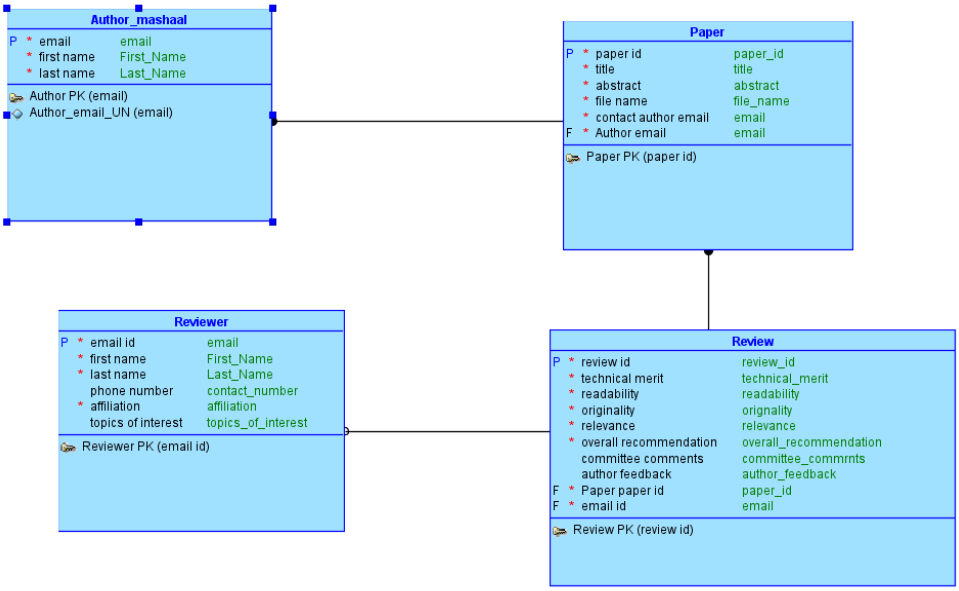

Patient_mashaal
P * PatientId VARCHAR2 (10)
* First_name VARCHAR2 (20)
* Last_name VARCHAR2 (20)
DtaeOfBirth DATE
Gender VARCHAR2 (7)
Contact_no NUMBER (2)
Address VARCHAR2 (25)
Patient_mashaal_PK (PatientId)
Medical_Record
P * Record_id VARCHAR2
* date DATE
diagnosis VARCHAR2
treatement VARCHAR2
medication VARCHAR2
F * Patient_mashaal_PatientId VARCHAR2
F * Doctor_Doctor_id VARCHAR2
Medical_Record_PK (Record_id)
Billing
P * billing_id VARCHAR2 (20)
* date DATE
* Total_amount NUMBER (2)
* payment_status VARCHAR2 (15)
F * Doctor_Doctor_id VARCHAR2 (15)
F * Patient_mashaal_PatientId VARCHAR2 (10)
Billing_PK (billing_id)
Billing_Doctor_FK (Doctor_Doctor_id)
Billing_Patient_mashaal_FK (Patient_mashaal_PatientId)
Appointment
P * appointmnet_id VARCHAR2 (20)
* appointment_date DATE
* appointment_time DATE
* appointment_status VARCHAR2 (20)
F * Patient_mashaal_PatientId VARCHAR2 (10)
F * Doctor_Doctor_id VARCHAR2 (15)
PF * Billing_billing_id VARCHAR2 (20)
Appointment_PK (appointmnet_id, Billing_billing_id)
Appointment_Billing_FK (Billing_billing_id)
Appointment_Patient_mashaal_FK (Patient_mashaal_PatientId)
Appointment_Doctor_FK (Doctor_Doctor_id)
Doctor
P * Doctor_id VARCHAR2 (15)
* First_name VARCHAR2 (20)
* Last_name VARCHAR2 (20)
* Specialization VARCHAR2 (20)
Contact_no NUMBER (2)
email VARCHAR2 (25)
Doctor_PK (Doctor_id)

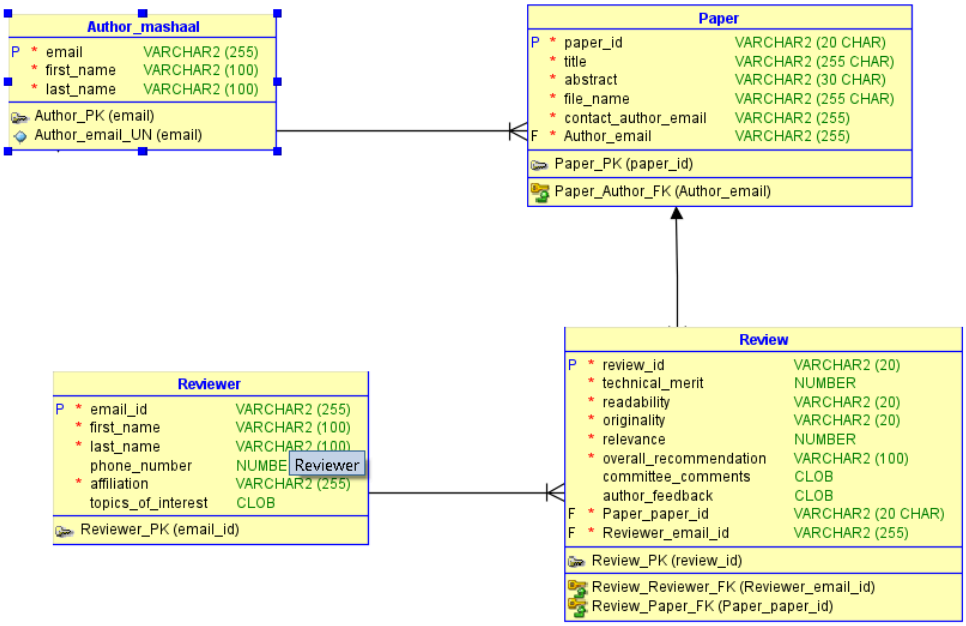
TASK#4

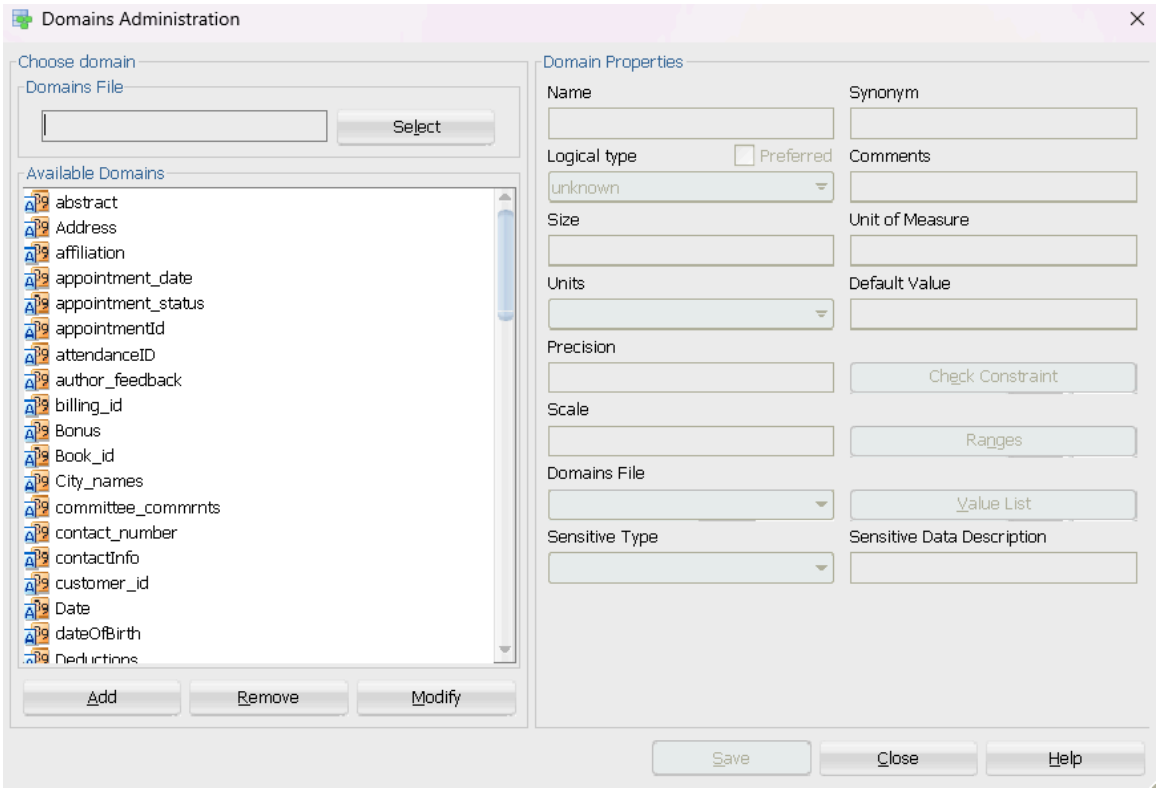

Department
P * DepartmentID Department_id
* Name Dep_name
* Location location
F * ManagerId managerID
F * Employee_id Employee_id
Department PK (DepartmentID)
Payroll
P * PayRoll_ID payroll_id
* Salary salary
* Bonus Bonus
* Deductions Deductions
* NetSalary NetSalary
F * ManagerId managerID
F * Employee_id Employee_id
Payroll PK (PayRoll_ID)
Employee_mashaal
P * Employee_id Employee_id
* Name Name
DOB dateOfBirth
Contact contact_number
U * Email email
* HireDate Hire_date
P * ManagerId managerID
F * DepartmentID Department_id
F * AttendanceID attendanceID
F * AttendanceID1 attendanceID
F * PayRoll_ID payroll_id
Employee_mashaal PK (ManagerId, Employee_id)
Project
P * ProjectID Product_id
* Name project_name
* StartDate Date
* EndDate Date
* Status status
F * ManagerId managerID
F * Employee_id Employee_id
Project PK (ProjectID)
Employee_project
P * EmpProjID ProjectID
* Role Role
Employee_project PK (EmpProjID)
Attendance
P * AttendanceID attendanceID
* Date Date
* Status status
F * ManagerId managerID
F * Employee_id Employee_id
Attendance PK (AttendanceID)

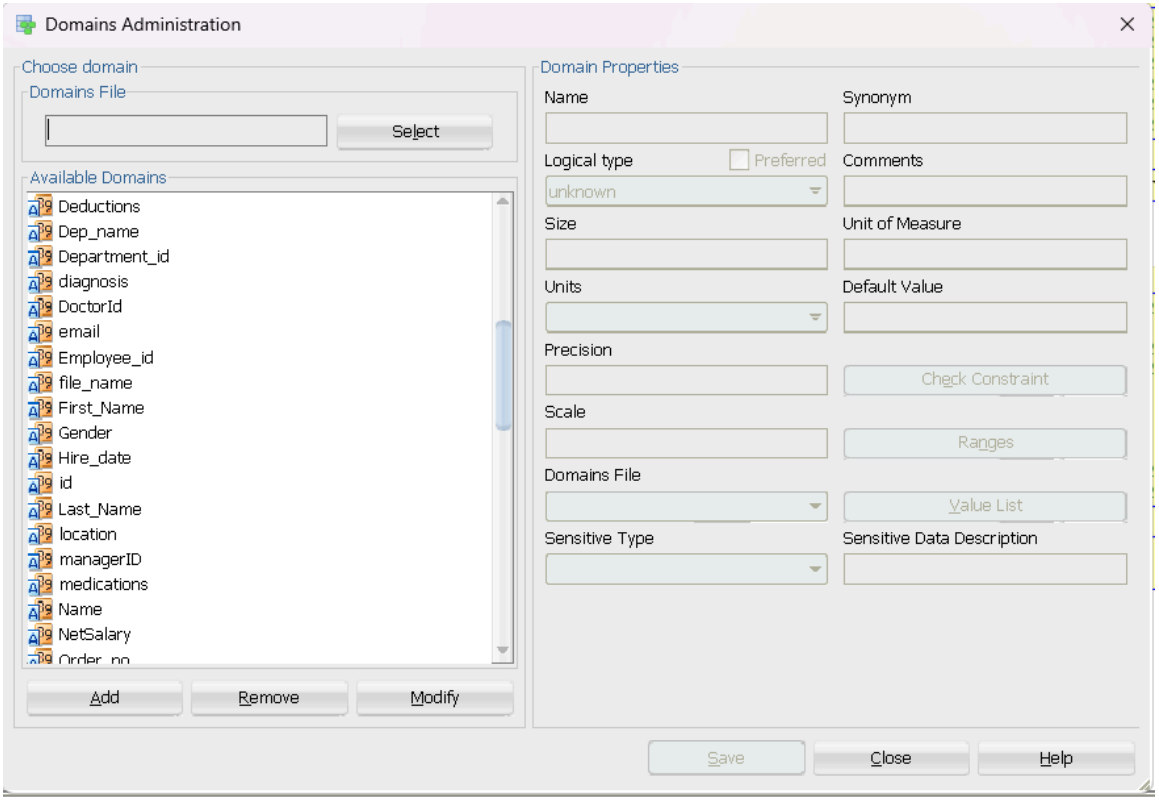

Department
Employee_mashaal
Payroll
Project
Attendance
involves
Employee_project

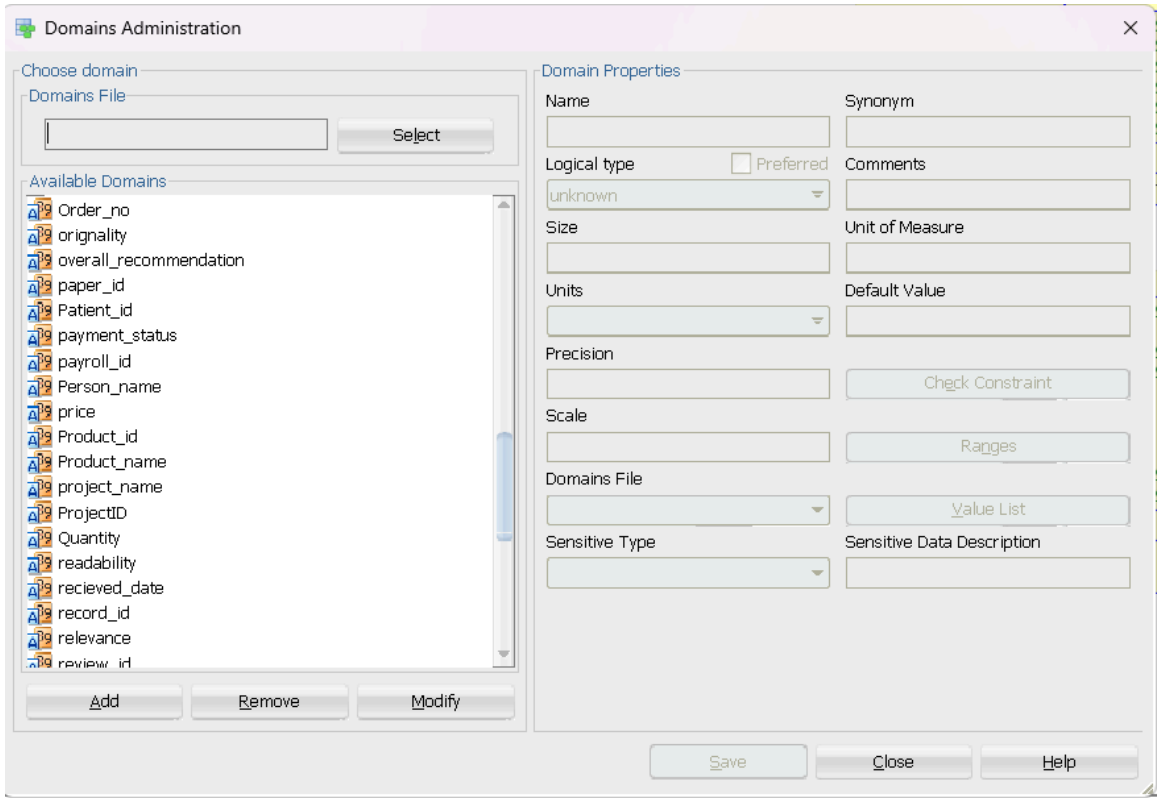
TASK#5

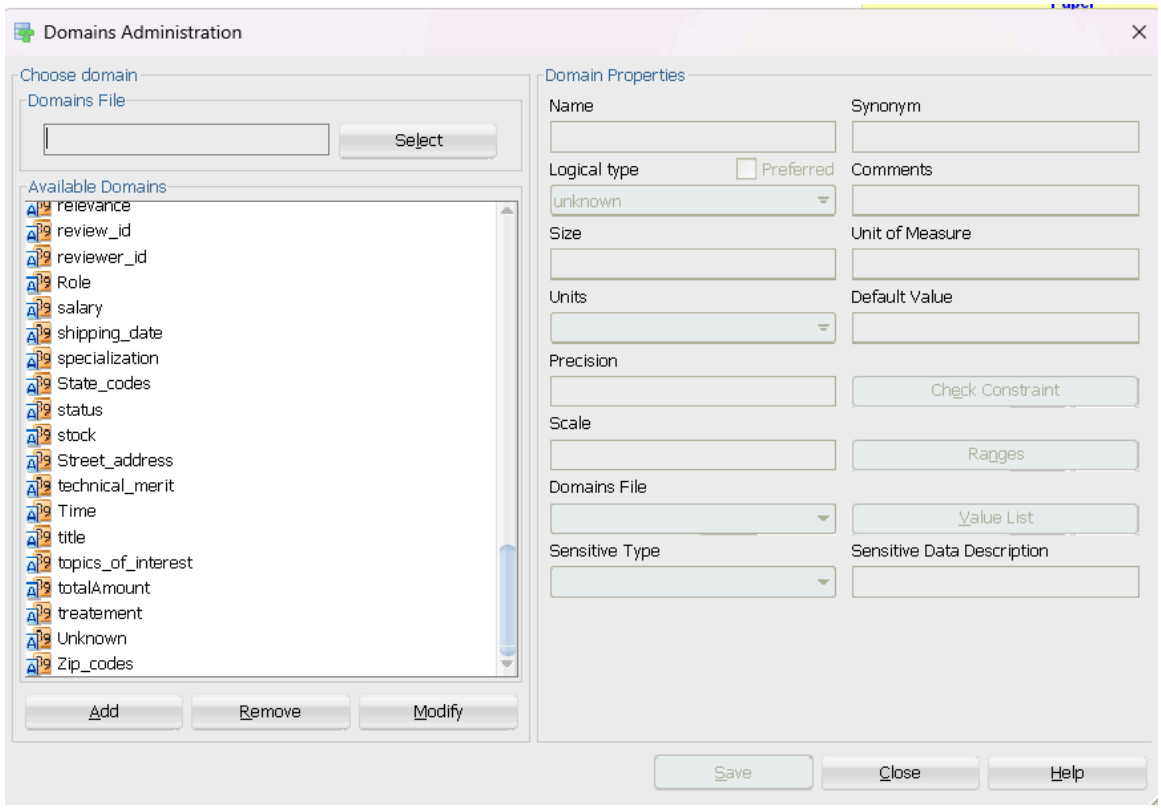

Author_mashaal
P * email email
* first name First_Name
* last name Last_Name
Author PK (email)
Author_email_UN (email)
Paper
P * paper id paper_id
* title title
* abstract abstract
* file name file_name
* contact author email email
F * Author email email
Paper PK (paper id)
Reviewer
P * email id email
* first name First_Name
* last name Last_Name
phone number contact_number
* affiliation affiliation
topics of interest topics_of_interest
Reviewer PK (email id)
Review
P * review id review_id
* technical merit technical_merit
* readability readability
* originality orignality
* relevance relevance
* overall recommendation overall_recommendation
committee comments committee_commrnts
author feedback author_feedback
F * Paper paper id paper_id
F * email id email
Review PK (review id)


Author_mashaal
P * email VARCHAR2 (255)
* first_name VARCHAR2 (100)
* last_name VARCHAR2 (100)
Author_PK (email)
Author_email_UN (email)
Paper
P * paper_id VARCHAR2 (20 CHAR)
* title VARCHAR2 (255 CHAR)
* abstract VARCHAR2 (30 CHAR)
* file_name VARCHAR2 (255 CHAR)
* contact_author_email VARCHAR2 (255)
F * Author_email VARCHAR2 (255)
Paper_PK (paper_id)
Paper_Author_FK (Author_email)
Reviewer
P * email_id VARCHAR2 (255)
* first_name VARCHAR2 (100)
* last_name VARCHAR2 (100)
phone_number NUMBE
Reviewer
* affiliation VARCHAR2 (255)
topics_of_interest CLOB
Reviewer_PK (email_id)
Review
P * review_id VARCHAR2 (20)
* technical_merit NUMBER
* readability VARCHAR2 (20)
* originality VARCHAR2 (20)
* relevance NUMBER
* overall_recommendation VARCHAR2 (100)
committee_comments CLOB
author_feedback CLOB
F * Paper_paper_id VARCHAR2 (20 CHAR)
F * Reviewer_email_id VARCHAR2 (255)
Review_PK (review_id)
Review_Reviewer_FK (Reviewer_email_id)
Review_Paper_FK (Paper_paper_id)


Domains Administration
Choose domain
Domains File
Select
Available Domains
abstract
Address
affiliation
appointment_date
appointment_status
appointmentId
attendanceID
author_feedback
billing_id
Bonus
Book_id
City_names
committee_commrnts
contact_number
contactInfo
customer_id
Date
dateOfBirth
Add
Remove
Modify
Domain Properties
Name
Synonym
Logical type
Preferred
Comments
unknown
Size
Unit of Measure
Units
Default Value
Precision
Check Constraint
Scale
Ranges
Domains File
Value List
Sensitive Type
Sensitive Data Description
Save
Close
Help


Domains Administration
Choose domain
Domains File
Select
Available Domains
Deductions
Dep_name
Department_id
diagnosis
DoctorId
email
Employee_id
file_name
First_Name
Gender
Hire_date
id
Last_Name
location
managerID
medications
Name
NetSalary
Add
Remove
Modify
Domain Properties
Name
Synonym
Logical type
Preferred
Comments
unknown
Size
Unit of Measure
Units
Default Value
Precision
Check Constraint
Scale
Ranges
Domains File
Value List
Sensitive Type
Sensitive Data Description
Save
Close
Help


Domains Administration
Choose domain
Domains File
Select
Available Domains
Order_no
orignality
overall_recommendation
paper_id
Patient_id
payment_status
payroll_id
Person_name
price
Product_id
Product_name
project_name
ProjectID
Quantity
readability
recieved_date
record_id
relevance
review_id
Add
Remove
Modify
Domain Properties
Name
Synonym
Logical type
Preferred
Comments
unknown
Size
Unit of Measure
Units
Default Value
Precision
Check Constraint
Scale
Ranges
Domains File
Value List
Sensitive Type
Sensitive Data Description
Save
Close
Help


Domains Administration
Choose domain
Domains File
Select
Available Domains
relevance
review_id
reviewer_id
Role
salary
shipping_date
specialization
State_codes
status
stock
Street_address
technical_merit
Time
title
topics_of_interest
totalAmount
treatement
Unknown
Zip_codes
Add
Remove
Modify
Domain Properties
Name
Synonym
Logical type
Preferred
Comments
unknown
Size
Unit of Measure
Units
Default Value
Precision
Check Constraint
Scale
Ranges
Domains File
Value List
Sensitive Type
Sensitive Data Description
Save
Close
Help